MATH 180 Strategies of Problem Solving

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Outline

1. SoPS Lecture 17
   - Objects and statements
   - Describing sets
   - Subsets and the empty set
   - Sets containing sets
2. Set algebra
   - Union, intersection, and difference
   - Counting and power sets
   - Ordered pairs and products
   - Quantifiers and proofs
3. Why is it called naive?
   - Russell's paradox

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What are we learning to do?

Our questions

- What objects does an expression describe?
- Does a sentence make mathematical sense?
- If it makes sense, is it true?
- How can we justify our answer?

A false statement can still make perfectly good mathematical sense.

A first sorting puzzle

You know ordinary arithmetic. Classify each line.

$$2+3=5 \qquad 2+3=8 \qquad 2+3 \qquad 2+ \ =5$$

Four possible responses

- A meaningful true statement.
- A meaningful false statement.
- A meaningful expression, but no claim to evaluate.
- An incomplete or undefined expression.

A first sorting puzzle — discussion

- $2 + 3 = 5$ is a true statement.
- $2 + 3 = 8$ is a false statement.
- $2 + 3$ names a number. It makes sense, but is not a statement.
- $2 + \ = 5$ is incomplete as written. We could repair it by introducing a variable.

One more possibility

$x + 3 = 5$ is an **open sentence**. Its truth depends on x.

Sets

Working description

A **set** is a collection of distinct objects, called its **elements** or **members**.

- $A = \{1, 2, 3\}$ is a set of numbers.
- $V = \{a, e, i, o, u\}$ is a set of letters.
- A set can contain game positions, people, or other sets.

We must be able to determine what counts as an element.

Membership

Let $A = \{1, 2, 3\}$.

$2 \in A$ means that 2 is an element of A.

$5 \notin A$ means that 5 is not an element of A.

Reading aloud

Read $\in$ as "is an element of." This helps distinguish membership from other relationships.

Which lines make sense?

Let $A = \{1, 2, 3\}$. We use numbers as ordinary numbers.

Discuss with a partner

- $2 \in A$
- $5 \in A$
- $A + 2 = 5$
- A

Which lines make sense? — discussion

Let $A = \{1, 2, 3\}$. We use numbers as ordinary numbers.

- $2 \in A$: meaningful and true.
- $5 \in A$: meaningful and false.
- $A + 2 = 5$: we have not defined addition of a set and a number.
- A: a meaningful name for a set, but no statement.

Conventions matter

Our convention

Numbers and letters are ordinary objects. We do not use a special set-theoretic encoding of numbers.

- Expressions like $2 \subseteq A$ will be undefined in our language, because 2 is not being treated as a set.
- Set membership $x \in A$ requires A to be a set. The object x may itself be a set.
- An unfamiliar but well-defined expression may be false. Unfamiliarity alone does not make it meaningless.

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Order and repetition

$$\{1, 2, 3\} = \{3, 1, 2\} = \{1, 1, 2, 3, 3\}.$$

Equality of sets

Two sets are equal exactly when they have the same elements.

- Order does not matter.
- Listing an element again does not create another element.

A set of dice outcomes and a sequence of dice rolls record different information.

The same set?

For each proposed equality, decide whether it is true.

Discuss with a partner

- $\{2, 4, 6\} = \{6, 2, 4\}$
- $\{1, 1, 2\} = \{1, 2, 2\}$
- $\{1, 2\} = \{3\}$
- $\{1, 2\} = 3$

The same set? — discussion

For each proposed equality, decide whether it is true.

- True: the elements are identical.
- True: each side has exactly the elements 1 and 2.
- False: 1 belongs to the left side and not the right.
- False under our convention: a two-element set is not the number 3. Equality can compare objects of different kinds.

Set-builder notation

$$E = \{x \in \mathbb{Z} : x \text{ is even}\}.$$

- $x \in \mathbb{Z}$ specifies the allowed objects.
- The colon means "such that.
- The condition selects which allowed objects belong.

Finite example

$\{x \in \mathbb{Z} : 0 < x < 5\} = \{1, 2, 3, 4\}$.

Some familiar infinite sets

$$\mathbb{N} = \{1, 2, 3, \ldots\},$$
$$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\},$$
$$\mathbb{Q} = \{a/b : a, b \in \mathbb{Z},\ b \neq 0\},$$
$$\mathbb{R} = \text{the set of real numbers.}$$

In these slides, $0 \notin \mathbb{N}$. Some books use a different convention.

Infinite sets need a description that specifies every element.

Reading a set description

Let $B = \{x \in \mathbb{Z} : x^2 < 10\}$.

Discuss with a partner

- List all elements of B.
- Is $\sqrt{2} \in B$?
- Is $-3 \in B$?
- Is $\{x \in \mathbb{Z} : x^2\}$ a complete set description?

Reading a set description — discussion

Let $B = \{x \in \mathbb{Z} : x^2 < 10\}$.

- $B = \{-3, -2, -1, 0, 1, 2, 3\}$.
- No. $\sqrt{2}$ fails the requirement $x \in \mathbb{Z}$.
- Yes, because $(-3)^2 = 9 < 10$.
- No. x^2 is a number, not a condition to test.

Conditions versus output expressions

$$\{x \in \mathbb{Z} : x^2 < 10\} \qquad \{x^2 : x \in \mathbb{Z}\}$$

- The first selects integers satisfying a condition.
- The second collects the outputs x^2 for all integers x.
- The second set is $\{0, 1, 4, 9, 16, \ldots\}$.

Check

Why does the second set contain 9 only once, even though $3^2 = (-3)^2$?

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Subsets

Definition

$A \subseteq B$ means that **every element of A is also an element of B.**

$$\{1, 3\} \subseteq \{1, 2, 3\}.$$

- $A \subseteq A$ for every set A.
- $A \subsetneq B$ means $A \subseteq B$ and $A \neq B$.
- We use $\subseteq$ to allow equality and $\subsetneq$ to exclude it.

The symbol $\subset$ has different conventions in different books.

Membership versus inclusion

Let $A = \{1, 2, 3\}$.

$$2 \in A \qquad \{2\} \subseteq A.$$

What are we comparing?

- $x \in A$ asks whether x is one of the elements of A.
- $B \subseteq A$ asks whether every element of B belongs to A.

$\{2\}$ is a set with one element. Its element is 2.

Membership or subset?

Let $A = \{1, 2, 3\}$. Classify each line.

Discuss with a partner

- $1 \in A$ and $\{1\} \in A$
- $\{1\} \subseteq A$ and $A \subseteq A$
- $1 \subseteq A$
- $A \in A$

Membership or subset? — discussion

Let $A = \{1, 2, 3\}$. Classify each line.

- True and false, respectively. The elements of A are numbers.
- Both true.
- Undefined in our convention: the left side must be a set.
- Meaningful and false. None of the listed elements is A.

The empty set

Definition

The **empty set**, written $\varnothing$ or $\{\}$, has no elements.

$$\{x \in \mathbb{R} : x^2 = -1\} = \varnothing.$$

- $\varnothing$ is a set.
- $\{\varnothing\}$ has one element: the empty set.
- $\varnothing \neq \{\varnothing\}$.

Question

Is a box containing one empty box itself empty?

Why the empty set is a subset of every set

For any set A,

$$\varnothing \subseteq A.$$

Reason

To disprove this, we would need an element of $\varnothing$ that fails to belong to A. There is no element of $\varnothing$ at all.

This is an example of **vacuous truth**.

"Every dragon in this room wears a hat is true if there are no dragons in this room.

Empty-set check

Let $A = \{1, 2\}$.

Discuss with a partner

- $\varnothing \subseteq A$
- $\varnothing \in A$
- $\varnothing \in \{\varnothing\}$
- $\{\varnothing\} \subseteq A$

Empty-set check — discussion

Let $A = \{1, 2\}$.

- True: the empty set is a subset of every set.
- False: the only elements of A are 1 and 2.
- True: it is the one listed element.
- False: this would require $\varnothing \in A$.

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Sets can be elements

Let

$$C = \{1, \{2, 3\}, \varnothing\}.$$

- The elements of C are 1, $\{2, 3\}$, and $\varnothing$.
- $\{2, 3\} \in C$ is true.
- $2 \in C$ is false.
- $\{2, 3\} \subseteq C$ is false.

Braces are structure

A set containing a set does not automatically contain the elements of that inner set.

Braces change the answer

Let $D = \{1, \{1\}, \{2, 3\}\}$.

Discuss with a partner

- $1 \in D$ and $\{1\} \in D$
- $\{1\} \subseteq D$
- $\{\{1\}\} \subseteq D$
- $\{2, 3\} \subseteq D$

Braces change the answer — discussion

Let $D = \{1, \{1\}, \{2, 3\}\}$.

- Both true: both objects occur in the list.
- True: its only element, 1, belongs to D.
- True: its only element, $\{1\}$, belongs to D.
- False: neither 2 nor 3 is an element of D.

A relation between the two symbols

Always true

For every object x and every set A,

$$x \in A \iff \{x\} \subseteq A.$$

- The only element of $\{x\}$ is x.
- So checking every element of $\{x\}$ means checking x.

Careful

This does not say $x \in A$ is equivalent to $\{x\} \in A$.

A short group challenge

Construct a set A satisfying each request separately.

- $1 \in A$ but $\{1\} \notin A$.
- $\{1\} \in A$ but $1 \notin A$.
- Both $1 \in A$ and $\{1\} \in A$.
- $\varnothing \in A$ and A has exactly two elements.

Compare constructions. Can different groups give different correct answers?

A short group challenge — examples

- $A = \{1\}$.
- $A = \{\{1\}\}$.
- $A = \{1, \{1\}\}$.
- $A = \{\varnothing, 7\}$.

Another check

Why does $\{\varnothing, \varnothing\}$ fail?

It has only one element. Repetition does not count.

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Union and intersection

For sets A and B,

$$A \cup B = \{x : x \in A \text{ or } x \in B\},$$
$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

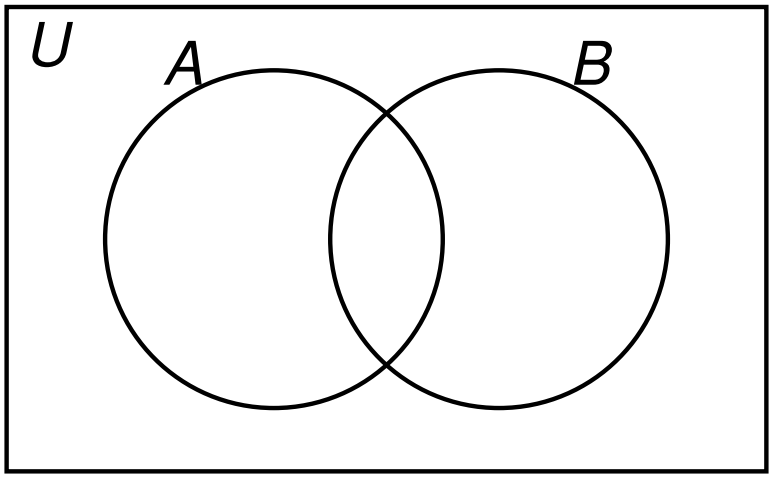
The mathematical word or

An element belongs to $A \cup B$ if it belongs to either set **or to both**.

Example: if $A = \{1, 2, 3\}$ and $B = \{3, 4\}$,

$$A \cup B = \{1, 2, 3, 4\}, \qquad A \cap B = \{3\}.$$

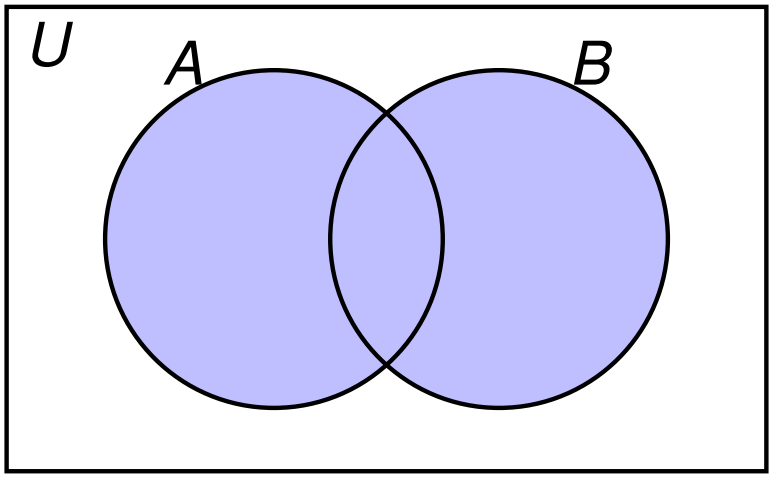
Reading a Venn diagram



The rectangle is the universe U. The circles represent sets A and B.

Each position represents a possible element. The overlap represents elements belonging to both sets.

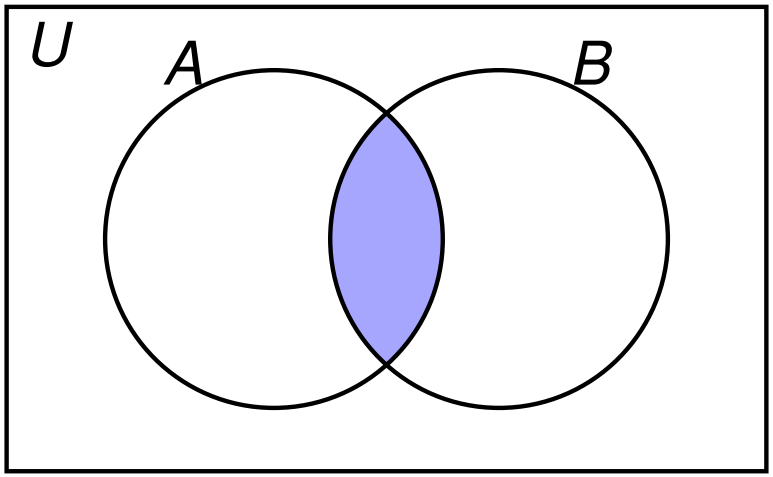
Union in a Venn diagram



$$x \in A \cup B \iff x \in A \text{ or } x \in B.$$

The shaded region includes the overlap. Mathematical “or” allows both.

Intersection in a Venn diagram



$$x \in A \cap B \iff x \in A \text{ and } x \in B.$$

Check: Does a point in A but outside B belong to the shaded set?

Difference and complement

$$A \setminus B = \{x \in A : x \notin B\}.$$

If $A = \{1, 2, 3\}$ and $B = \{3, 4\}$,

$$A \setminus B = \{1, 2\}, \qquad B \setminus A = \{4\}.$$

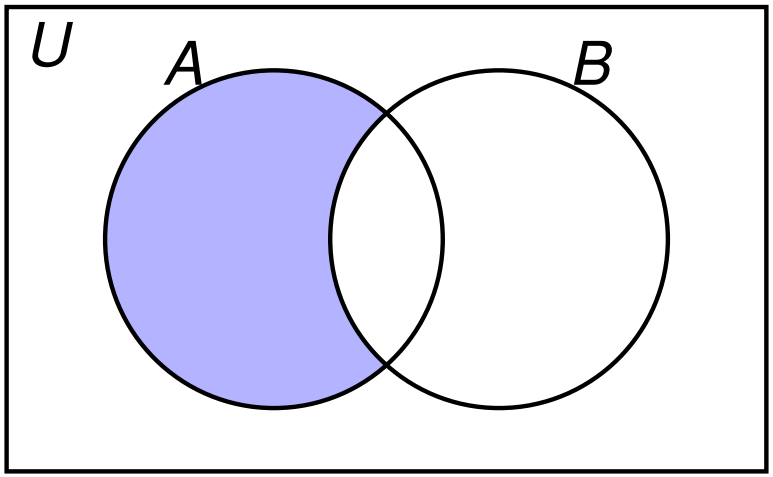
Complement needs a universe

If all objects under discussion lie in a fixed set U, then

$$A^c = U \setminus A.$$

Without a specified or understood U, the complement is ambiguous.

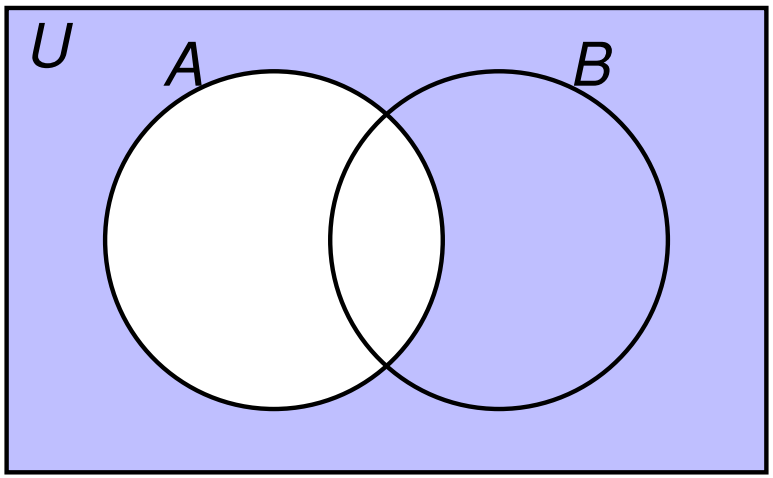
Difference in a Venn diagram



$$A \setminus B = \{x \in A : x \notin B\}.$$

Check: Where would we shade $B \setminus A$ instead?

Complement in a Venn diagram



The shaded region is $A^c = U \setminus A$.

Check: Why does part of B belong to A^c, while another part does not?

Calculating with sets

Let $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 4\}$, $B = \{2, 3, 4\}$.

Discuss with a partner

- Find $A \cup B$ and $A \cap B$.
- Find $A \setminus B$ and $B \setminus A$.
- Find A^c.
- Is $A \cap B = 2$ true?

Calculating with sets — discussion

Let $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 4\}$, $B = \{2, 3, 4\}$.

- $A \cup B = \{1, 2, 3, 4\}$ and $A \cap B = \{2, 4\}$.
- $A \setminus B = \{1\}$ and $B \setminus A = \{3\}$.
- $A^c = \{3, 5, 6\}$.
- False. The intersection is a set, not the number of its elements.

Which expressions mean something?

Let $A = \{1, 2\}$ and $B = \{2, 3\}$.

Discuss with a partner

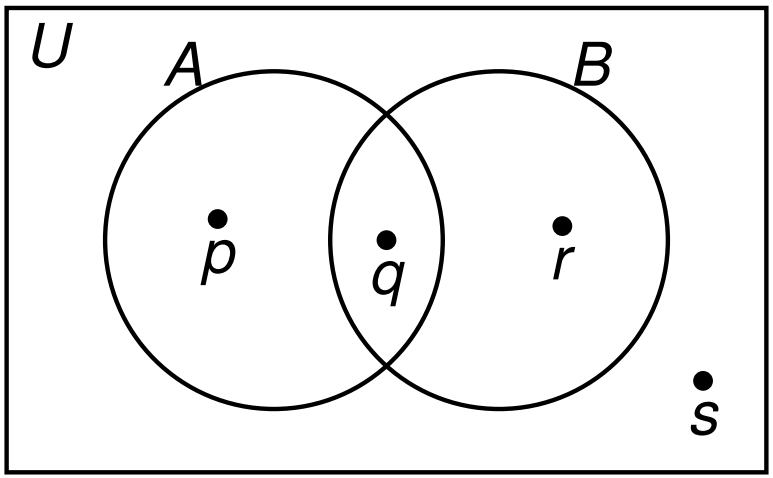
- $A \cup B$
- $A \cup 2$
- $2 \in A \cap B$
- $A \in B \cup \{A\}$

Which expressions mean something? — discussion

Let $A = \{1, 2\}$ and $B = \{2, 3\}$.

- Meaningful set expression: $\{1, 2, 3\}$. No truth value by itself.
- Undefined: union requires two sets in our convention.
- Meaningful and true, since $A \cap B = \{2\}$.
- Meaningful and true. The union has A as an element.

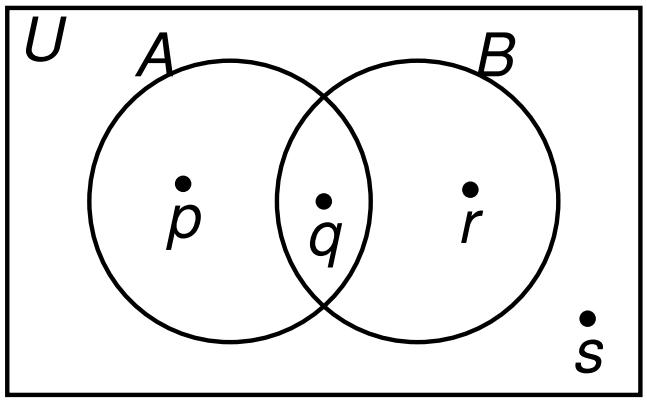
Venn diagram check



Which points belong to $A \cup B$, $A \cap B$, $A \setminus B$, and $(A \cup B)^c$?

Are $p \in A$ and $p \subseteq A$ equally meaningful? Treat p as a point, not a set.

Venn diagram check — answers



Among the labeled points: p, q, r belong to $A \cup B$, q belongs to $A \cap B$, p belongs to $A \setminus B$, and s belongs to $(A \cup B)^c$.

$p \in A$ is true. We have not defined $p \subseteq A$ for a point.

Disjoint sets

Definition

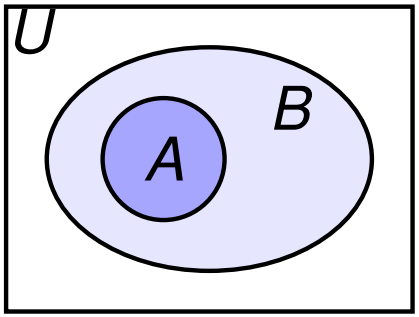
Sets A and B are **disjoint** if $A \cap B = \varnothing$.

- $\{1, 3, 5\}$ and $\{2, 4, 6\}$ are disjoint.
- $\{1, 2\}$ and $\{2, 3\}$ are different but not disjoint.
- $\{\varnothing\}$ and $\varnothing$ are disjoint.

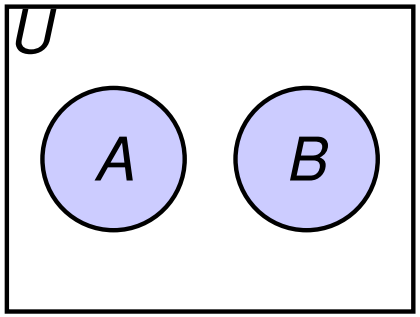
Check

Can a nonempty set be disjoint from itself? Explain.

Inclusion and disjointness in pictures



$A \subseteq B$



$A \cap B = \varnothing$

The left picture says every element of A belongs to B.

It does not assert $A \in B$. Circles contain points representing elements, not automatically the circles themselves.

Identities worth testing

For arbitrary sets A and B, which are always true?

- $A \cup B = B \cup A$.
- $A \cap B = B \cap A$.
- $A \setminus B = B \setminus A$.
- $A \cup \varnothing = A$ and $A \cap \varnothing = \varnothing$.

For a false claim, find a small counterexample.

Identities worth testing — discussion

- Union and intersection are commutative: the first two are true.
- Difference need not be commutative. Take $A = \{1\}$ and $B = \{2\}$. Then $A \setminus B = \{1\}$ and $B \setminus A = \{2\}$.
- Both empty-set identities are true.

A counterexample

One example where a universal claim fails disproves that claim. Many successful examples do not prove it.

De Morgan's laws

With a fixed universe U,

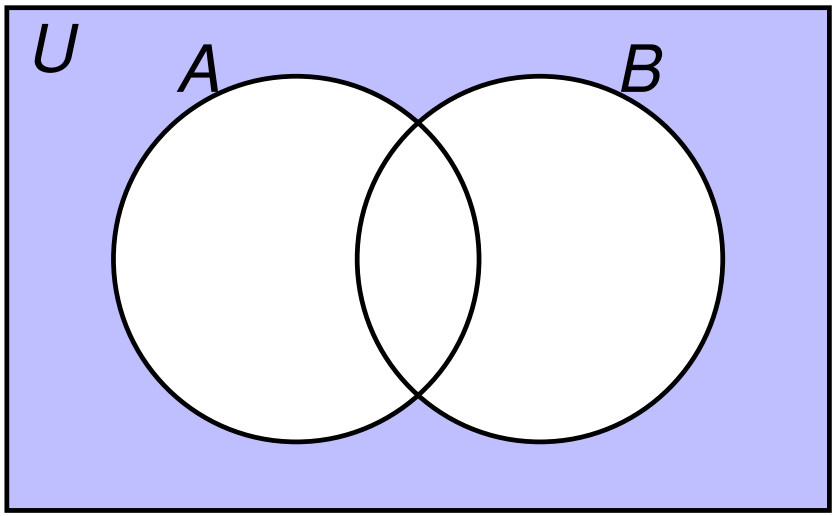
$$(A \cup B)^c = A^c \cap B^c, \qquad (A \cap B)^c = A^c \cup B^c.$$

Example in words

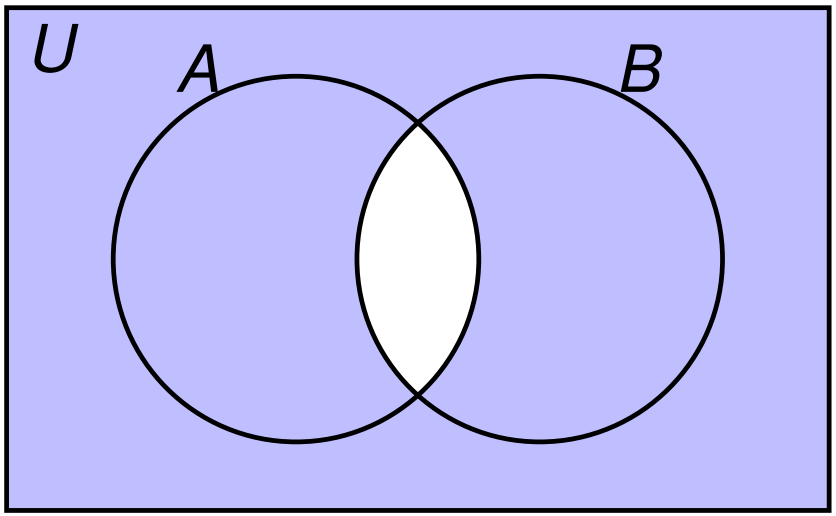
"Not (plays chess or plays Go) means "does not play chess and does not play Go.

Test both identities with $U = \{1, 2, 3, 4\}$, $A = \{1, 2\}$, and $B = \{2, 3\}$.

De Morgan's laws in pictures



$$(A \cup B)^c = A^c \cap B^c$$



$$(A \cap B)^c = A^c \cup B^c$$

Left: outside both circles.
Right: everywhere except the overlap.

Check: In the right picture, why is the part belonging only to A shaded?

De Morgan's laws — example

Here $A^c = \{3, 4\}$ and $B^c = \{1, 4\}$.

$$\begin{aligned}(A \cup B)^c &= \{1, 2, 3\}^c = \{4\},\\ A^c \cap B^c &= \{3, 4\} \cap \{1, 4\} = \{4\},\\ (A \cap B)^c &= \{2\}^c = \{1, 3, 4\},\\ A^c \cup B^c &= \{3, 4\} \cup \{1, 4\} = \{1, 3, 4\}.\end{aligned}$$

This verifies this example. A proof must work for every x and every choice of A, B.

Final thoughts

Reflection

- How do $x \in A$ and $\{x\} \subseteq A$ relate?
- Why is $\varnothing \subseteq A$ always true?
- Give a meaningful false statement and an undefined expression.
- Find $\mathcal{P}(\{\varnothing, 1\})$.

Submit your responses by **Tonight!**